SUDIP SHARMA

PhD Researcher in CO_2 Capture & Process Systems Engineering

sudip.sharma@ntnu.no | Trondheim, Norway | sudipsharma | 0000-0003-4049-6031

PROFILE

PhD researcher in process systems engineering at NTNU, building open-source first-principles models of next-generation CO_2 capture and coupling them with machine-learning surrogates and multi-objective optimisation. Backed by three years of industrial process R&D and quality-control experience and an Erasmus Mundus MSc in techno-economic analysis, life-cycle assessment and analytical chemistry.

EDUCATION

PhD (Electrification of Chemical Industries)

Norwegian University of Science and Technology (NTNU)

Oct 2024 – Sep 2027 | Trondheim, Norway

Supervisor: Prof. Thomas Alan Adams II (Process and Power group). Modelling, simulation and optimisation of electrified CO_2-capture processes in the open-source python framework, with a focus on Magnetic Induction Swing Adsorption (MISA).

Erasmus Mundus M.Sc. in Biological & Chemical Engineering for Sustainable Bioeconomy (BIOCEB)

URCA (FR), TalTech (EE), ULiège (BE), TU Graz (AT)

Sep 2022 – Aug 2024 | Europe

Thesis (defended at TU Graz under Prof. V. Hacker): coupled techno-economic & life-cycle assessment of polyaniline-coated Pt/C catalysts for PEMFC electrodes. Recycled-Pt sourcing cut cradle-to-gate global-warming impact by about 79% (47.2 to 9.8 kg CO_2-eq per g catalyst) vs. South-African Pt.

B.E. in Chemical Engineering

Kathmandu University (KU)

Aug 2015 – Oct 2019 | Kathmandu, Nepal

EXPERIENCE

Research Assistant (M.Sc. Thesis)

Graz University of Technology (TU Graz)

Jan 2024 – Sep 2024 | Graz, Austria

- Built a coupled TEA + cradle-to-gate LCA (OpenLCA, ecoinvent v3.10, ReCiPe 2016) of two PANI-coated Pt/C production routes for PEMFC MEAs
- Found the Pt precursor dominates cost (about 33% of about € 59/g) and that recycled-Pt sourcing cuts global-warming impact by about 79% vs. primary South-African Pt

DOCTORAL RESEARCH

Magnetic Induction Swing Adsorption for CO_2 Capture

NTNU, Process and Power

Oct 2024 – Present | Trondheim

- Developed a validated open-source MISA model in IDAES/Pyomo, extending the `FixedBed1D` class with Specific Absorption Rate based electromagnetic-heating physics for an Fe_3O_4@HKUST-1 adsorbent
- Fitted and validated against adsorbent isotherm, breakthrough and desorption data: breakthrough within 9%, peak temperature within 5%
- Gaussian-process surrogates with multi-objective optimisation (purity, recovery, productivity, energy) for cycle design and TSA/PSA techno-economic comparison
- Funded by the NTNU–MIT Energy Research Programme (NMERP-C2)

PUBLICATIONS

Sharma, **Sudip** and Thomas A. Adams II (Oct. 2026). "Cycle Design and Surrogate-Based Multi-Objective Optimisation of Magnetic Induction Swing Adsorption for Electrified Post-Combustion CO_2 Capture". In: *Chemical Engineering Research and Design* 234, pp. 720–739. DOI: 10.1016/j.cherd.2026.09.029.

Sharma, **Sudip** and Thomas A. Adams II (June 2026). "An Open-Source IDAES Framework for Simulating Inductively Heated Adsorption Processes". In: *Systems and Control Transactions* 5. ESCAPE-36, Sheffield, UK, pp. 802–809. DOI: 10.69997/sct.177673.

Kafle, Saroj Raj, Sagar Ban, Rakesh Shrestha, Sijan Devkota, **Sudip Sharma**, Anup Kaini, and Ananda Tiwari (July 2026). "Recent Developments on Water Electrolyzers for Hydrogen Production: A Technological Review". In: *International Journal of Energy Research* 2026.1, 7663279. DOI: 10.1155/er/7663279.

Shrestha, Rakesh, Sagar Ban, Sijan Devkota, **Sudip Sharma**, Rajendra Joshi, Arjun Prasad Tiwari, Hak Yong Kim, and Mahesh Kumar Joshi (Aug. 2021). "Technological trends in heavy metals removal from industrial wastewater: A review". In: *Journal of Environmental Chemical Engineering* 9.4, 105688. DOI: 10.1016/j.jece.2021.105688.

TECHNICAL SKILLS

IDAES & Pyomo | Python | Gaussian-Process Surrogates | Active Learning | Multi-objective Optimisation

- Ran electrocatalysis characterisation (RDE, SEM, FTIR) of the Pt/C catalyst

Executive Engineer (R&D and Quality Control)

Jawalakhel Group of Industries

2019 – 2022 | Lalitpur, Nepal

- Ran quality control and analytical testing of raw materials and finished products
- Supported process R&D and product development on the plant floor

CONFERENCE CONTRIBUTIONS

GHGT-18 – Student Poster Award

18th IEAGHG Greenhouse Gas Control Technologies Conf.

25–29 Oct 2026 | Perth, Australia

E-poster and lightning talk: multi-objective optimisation of MISA for electrified CO_2 capture.

ESCAPE-36

European Symposium on Computer Aided Process Engineering

21–24 Jun 2026 | Sheffield, UK

First-author paper (*Syst. Control Trans.* 5): open-source IDAES MISA model validated against experimental isotherm, breakthrough and desorption data (breakthrough within 9%, peak bed temperature within 5%).

CHISA 2026

28th Int. Congress of Chemical & Process Engineering

23–27 Aug 2026 | Prague, Czech Republic

Poster presentation on the MISA process model.

SIMS 2026

67th Conf. of the Scandinavian Simulation Society

16–17 Sep 2026 | Eskilstuna, Sweden

Discussion paper on the MISA process model.

Process Modelling & Simulation | TEA & LCA | OpenLCA & ecoinvent | HPLC | FTIR | GC-MS

AWARDS

Student Poster Award
IEAGHG, GHGT-18, Perth (2026)

Erasmus Mundus Scholarship
European Union (2022–2024)

Diligent Student Scholarship
Kathmandu University (2016–2019)

LANGUAGES

Nepali (native) ●●●●●

English ●●●●●

Norwegian ●●◐○○

LEADERSHIP

Founding Secretary
KU Circle of Noble Chemineers (KUCONC)

Organised technical seminars and project demonstrations for 50+ chemical engineering students.

REFERENCES

Prof. Thomas Alan Adams II
@ NTNU
thomas.a.adams@ntnu.no

Prof. Viktor Hacker
@ TU Graz
viktor.hacker@tugraz.at